

CS228 : Logic for Computer Science

Module 1: Propositional Logic

Lecture 4: Natural Deduction Rules

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Quiz 1

- ▶ Aug 28th 5.10pm
- ▶ Venue: Most likely LH 101, 102, LT 105 (More details on Piazza)
- ▶ Syllabus: All material covered till Aug 25th.

Problemsheets and help sessions

- ▶ First problemsheet is out!
- ▶ Tomorrow 3.30pm there is an optional help session at CC101, 103, 105.

Recap: Lectures 1 and 2

Lecture 1: Syntax

- ▶ Atomic propositions; logical connectives ($\neg, \wedge, \vee, \rightarrow, \leftrightarrow, \oplus, \top, \perp$)
- ▶ Propositional formulas (defined inductively)

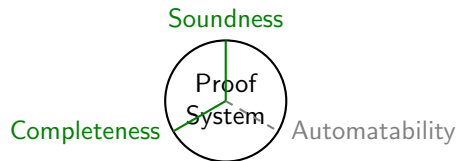
Lecture 2: Parse Trees, Semantics & Problems of Interest

- ▶ Parse trees / DAGs for formulas
- ▶ Semantics via truth tables
- ▶ Satisfiability, validity, semantic entailment ($\models$)

Recap: Lecture 3

Towards Formal Proofs

- ▶ Semantic **equivalence** ($\equiv$); used it to simplify formulas
- ▶ Minimal/universal connectives
- ▶ Truth tables don't scale (2^{100} rows!)
- ▶ Need a **formal proof system**
 - ▶ **Sequents** $F_1, \dots, F_n \vdash G$ is valid if we can derive G from $F_1, \dots, F_n$ using the rules of the proof system
 - ▶ **Soundness**: $F_1, \dots, F_n \vdash G \text{ valid} \Rightarrow F_1, \dots, F_n \models G$
 - ▶ **Completeness**: $F_1, \dots, F_n \models G \Rightarrow F_1, \dots, F_n \vdash G \text{ valid}$
 - ▶ Started **Natural deduction**: saw the $\wedge$ rules



Topic 4.1

Natural Deduction Rules

A formal proof system: Natural deduction

Natural deduction rules (from Huth & Ryan's Book)

12 Axioms or rules:

1. Introducing and Eliminating $\wedge$
2. Introducing and Eliminating $\rightarrow$
3. Introducing and Eliminating $\vee$
4. Negation rules: Introducing, Eliminating and Double negation
5. Bot rule

4 Derived/inferred rules to make life easier!



Gerhard Gentzen (1909–1945)

Conjunction Rules

If my cat is brown and it is fat then my cat is brown and fat!

$$\frac{F \quad G}{F \wedge G} \wedge i$$

$$\frac{F \wedge G}{F} \wedge e$$

$$\frac{F \wedge G}{G} \wedge e$$

- ▶ Rules to Introduce and Eliminate $\wedge$
- ▶ **Exercise 4.1** Show that $p \wedge q, r \vdash q \wedge r$ is a valid sequent.

1. $p \wedge q$ premise
2. r premise
3. q $\wedge e$ 1
4. $q \wedge r$ $\wedge i$ 3,2

When clear we omit subscripts e_1, e_2 and just write e .

Double negation rules

If it isn't the case that I don't like chocolate, then I like chocolate!

$$\frac{\neg\neg F}{F} \neg\neg e$$

$$\frac{G}{\neg\neg G} \neg\neg i$$

Exercise 4.2

Prove the sequent: $p, \neg\neg(q \wedge r) \vdash \neg\neg p \wedge r$ using only the rules till now!

1. p premise
2. $\neg\neg(q \wedge r)$ premise
3. $q \wedge r$ $\neg\neg e$ 2
4. r $\wedge e$ 3
5. $\neg\neg p$ $\neg\neg i$ 1
6. $\neg\neg p \wedge r$ $\wedge i$ 5,4

(Negation rules come later!)

Implication Rules

Whenever it rains, I carry an umbrella. It's raining — so I have my umbrella!

The **implies elimination rule** or **Modus Ponens MP**

$$\frac{F \quad F \rightarrow G}{G} \rightarrow e \text{ or MP}$$

Exercise 4.3: Show that $p, p \rightarrow q, p \rightarrow (q \rightarrow \neg\neg r) \vdash r$

- | | | |
|----|--|----------------|
| 1. | $p \rightarrow (q \rightarrow \neg\neg r)$ | premise |
| 2. | $p \rightarrow q$ | premise |
| 3. | p | premise |
| 4. | $q \rightarrow \neg\neg r$ | MP 1,3 |
| 5. | q | MP 2,3 |
| 6. | $\neg\neg r$ | MP 4,5 |
| 7. | r | $\neg\neg e$ 6 |

Implication Rules -2

If I were a cheetah then I would run fast. I cannot run fast, hence I am not a cheetah!

Another **implies elimination rule** or Modus Tollens MT

$$\frac{F \rightarrow G \quad \neg G}{\neg F} MT$$

Exercise 4.4

Show the following sequents are valid

1. $\neg p \rightarrow q, \neg q \vdash p$
2. $p \rightarrow \neg q, q \vdash \neg p$

1. $p \rightarrow \neg q$ premise
2. q premise
3. $\neg\neg q$ $\neg\neg i$ 2
4. $\neg p$ MT 1,3

Implication Rules -3

- ▶ By MT, we have $p \rightarrow q, \neg q \vdash \neg p$.
- ▶ This looks like the contrapositive! i.e., can we prove $p \rightarrow q \vdash \neg q \rightarrow \neg p$?
- ▶ But we don't know how to introduce $\rightarrow$ yet!
- ▶ Given $p \rightarrow q$, let us **assume** $\neg q$. Can we then prove $\neg p$?
- ▶ Yes, from premises AND assumption, using MT we get:

1.	$p \rightarrow q$	premise
2.	$\neg q$	assumption
3.	$\neg p$	MT 1,2
4.	$\neg q \rightarrow \neg p$	$\rightarrow i$ 2-3

Implication Rules -3 (Contd)

To convince someone that “if it rains, I’ll get wet,” you don’t check every possible day — you just imagine it raining, and show that you’d get wet!

$$\frac{\boxed{\begin{array}{c} F \\ \vdots \\ G \end{array}}}{F \rightarrow G} \rightarrow i$$

- ▶ Can use F and any premises so far; Can nest boxes, have multiple boxes etc.
- ▶ Line immediately after a closed box must match the pattern!
- ▶ F and G can be same! Consider premise p which shows $p \vdash p$, then

1.

p	assumption
-----	------------
2. $p \rightarrow p \quad \rightarrow i \ 1-1$

- ▶ We write $\vdash p \rightarrow p$ to denote it doesn’t depend on any premises.
- ▶ Formula F such that the sequent $\vdash F$ is valid is called a **theorem**.

Implication Rules -3 (Contd)

In English: “If mammals breathe air, and non-mammals aren’t whales, then whales breathe air!”

Example: Using all the rules, show that $\vdash (q \rightarrow r) \rightarrow [(\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r)]$

1.	$q \rightarrow r$	assumption
2.	$\neg q \rightarrow \neg p$	assumption
3.	p	assumption
4.	$\neg \neg p$	$\neg \neg i$ 3
5.	$\neg \neg q$	MT 2,4
6.	q	$\neg \neg e$ 5
7.	r	MP 1,6
8.	$p \rightarrow r$	$\rightarrow i$ 3-7
9.	$(\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r)$	$\rightarrow i$ 2-8
10.	$(q \rightarrow r) \rightarrow [(\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r)]$	$\rightarrow i$ 1-9

Converts proof of $F_1, \dots, F_n \vdash G$ to $\vdash F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$

Remember the Pop Quiz?

Back in the Pop Quiz, we asked whether $\varphi = (p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ is valid — and we had to check it with a truth table.

Now we can **prove** it instead!

1.	$(p \rightarrow q) \wedge (q \rightarrow r)$	assumption
2.	$p \rightarrow q$	$\wedge e_1$ 1
3.	$q \rightarrow r$	$\wedge e_2$ 1
4.	p	assumption
5.	q	MP 2,4
6.	r	MP 3,5
7.	$p \rightarrow r$	$\rightarrow i$ 4-6
8.	$(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$	$\rightarrow i$ 1-7

Some exercises

Exercise 4.5

Consider the following sequent

$$p \wedge q \rightarrow r \vdash p \rightarrow (q \rightarrow r)$$

1. Is it a valid sequent?
2. Which of the steps in your derivation are deterministic?
3. Write down the “converse” of the sequent. Is it a valid sequent?

Exercise 4.6

Using not more than 10 lines, prove that the following sequents are valid.

1. $(p \wedge q) \wedge r \vdash p \wedge (q \wedge r)$
2. $\vdash (p \wedge q) \rightarrow q$
3. $p \rightarrow q \wedge r \vdash (p \rightarrow q) \wedge (p \rightarrow r)$

Exercise 4.5: Solution

$$p \wedge q \rightarrow r \vdash p \rightarrow (q \rightarrow r)$$

- | | | |
|----|-----------------------------------|---------------------|
| 1. | $p \wedge q \rightarrow r$ | premise |
| 2. | p | assumption |
| 3. | q | assumption |
| 4. | $p \wedge q$ | $\wedge i$ 2,3 |
| 5. | r | MP 1,4 |
| 6. | $q \rightarrow r$ | $\rightarrow i$ 3-5 |
| 7. | $p \rightarrow (q \rightarrow r)$ | $\rightarrow i$ 2-6 |

$$p \rightarrow (q \rightarrow r) \vdash p \wedge q \rightarrow r$$

- | | | |
|----|-----------------------------------|---------------------|
| 1. | $p \rightarrow (q \rightarrow r)$ | premise |
| 2. | $p \wedge q$ | assumption |
| 3. | p | $\wedge e_1$ 2 |
| 4. | q | $\wedge e_2$ 2 |
| 5. | $q \rightarrow r$ | MP 1,3 |
| 6. | r | MP 5,4 |
| 7. | $p \wedge q \rightarrow r$ | $\rightarrow i$ 2-6 |

Disjunction rules

If the ball is round then the ball is either round or green.

$$\frac{F}{F \vee G} \vee i_1$$

$$\frac{G}{F \vee G} \vee i_2$$

$$\frac{F \vee G \quad \boxed{\begin{array}{c} F \\ \vdots \\ H \end{array}} \quad \boxed{\begin{array}{c} G \\ \vdots \\ H \end{array}}}{H} \vee e$$

► If $F \vee G$ is true and from BOTH F and G we are able to obtain H , then we can deduce H

Whether it's sunny or cloudy, I'll go for a walk — either way, I'm walking!

Disjunction rules (Contd)

Example: $q \rightarrow r \vdash (p \vee q) \rightarrow (p \vee r)$

1.	$q \rightarrow r$	premise
2.	$p \vee q$	assumption
3.	p	assumption
4.	$p \vee r$	$\vee i_1$ 3
5.	q	assumption
6.	r	MP 1,5
7.	$p \vee r$	$\vee i_2$ 6
8.	$p \vee r$	$\vee e$ 2, 3-4, 5-7
9.	$(p \vee q) \rightarrow (p \vee r)$	$\rightarrow i$ 2-8